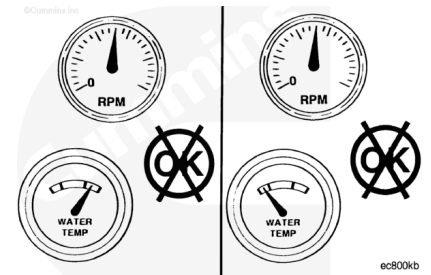


Trainings ▾

Leak Test

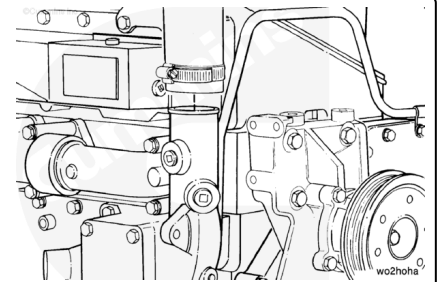
The engine thermostat and thermostat seal **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.

Refer to Procedure 008-016 in Section 8
(</qs3/pubsys2/xml/en/procedures/35/35-008-016-tr.html>) to inspect the thermostat seal.



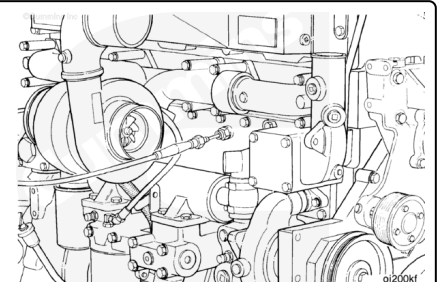
⚠ WARNING ⚠

Complete this test with the engine coolant temperature below 50°C [120°F]. Hot steam can cause serious personal injury.



Remove the radiator hose from the thermostat housing.

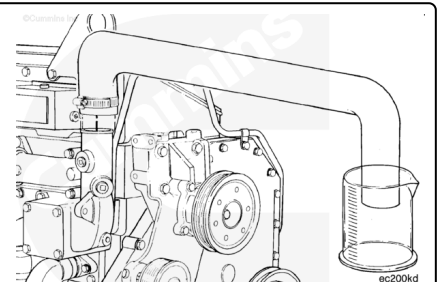
Install a thermocouple or temperature gauge, which is known to be accurate, in the water header plate or engine side of the thermostat housing.



Install a hose of the same size on the thermostat housing outlet. It **must** be long enough to reach a remote, dry container used to collect coolant.

Install and tighten a hose clamp on the housing outlet.

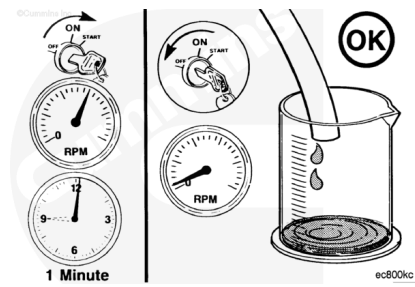
Place the other end of the hose in a dry container.



Operate the engine at rated rpm for 1 minute.

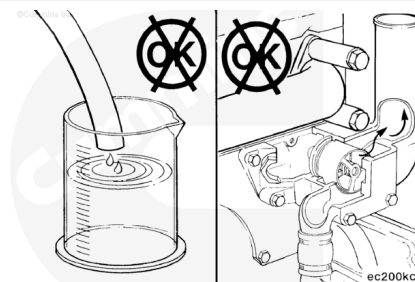
Shut off the engine, and measure the amount of coolant collected in the container.

The amount of coolant collected **must not** be more than 100 cc [3.3 fl oz].



If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat seal is leaking and **must** be replaced.

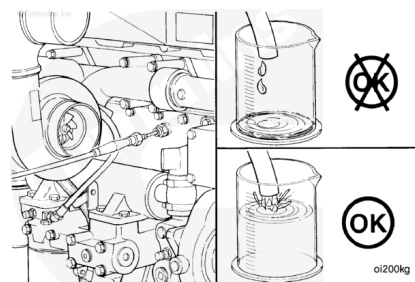
Refer to Procedure 008-016 in Section 8 (</qs3/pubsys2/xml/en/procedures/35/35-008-016-tr.html>) to replace the seal.



Complete the following test in-chassis to test the thermostat opening temperature.

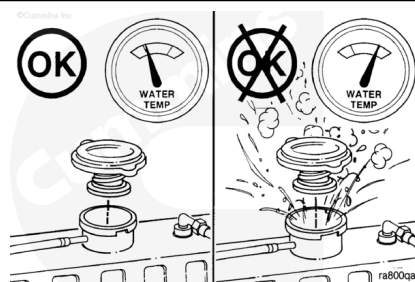
Start the engine and monitor the water temperature gauge and the container.

Thermostat Initial Opening Temperature		
celsius		fahrenheit
81	MIN	178
83	MAX	181



Shut off the engine when the coolant starts to flow.

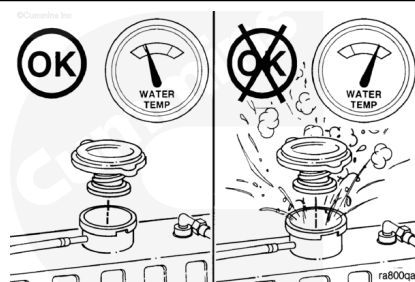
If coolant does **not** start flowing into the container during the initial opening temperature range, the thermostat **must** be replaced.



Remove

⚠ WARNING ⚠

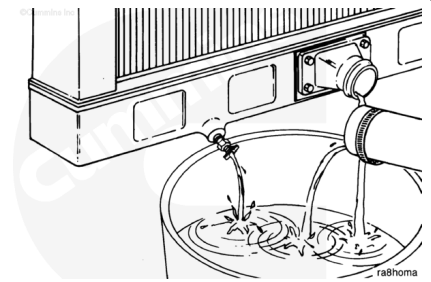
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the coolant system pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the pressure cap when the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



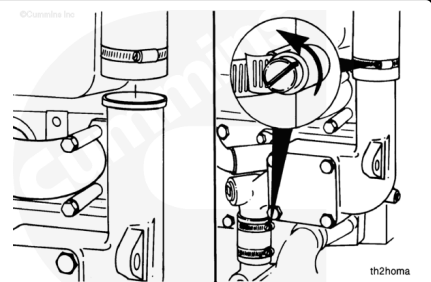
Drain the cooling system as follows:

- Open the radiator draincock.
- Remove the lower radiator hose.

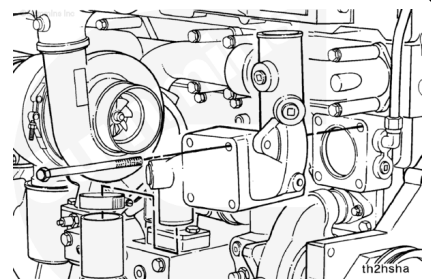
Remove the upper coolant hose from the thermostat housing.

Loosen the coolant bypass hose clamps.

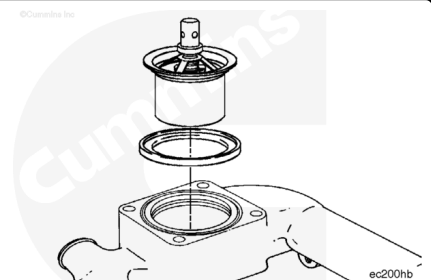
Note : Some models could have a converter cooler disc located in the bypass hose. Refer to Procedure 008-065 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-065-tr.html)



Remove the four thermostat housing mounting capscrews and the thermostat housing.



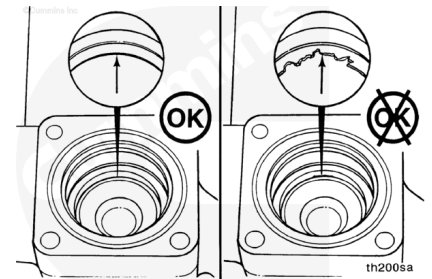
Remove the thermostat from the housing.



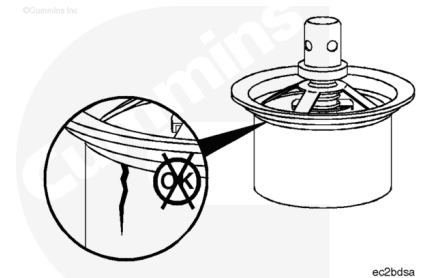
Inspect for Reuse

Inspect the thermostat seal for cracks, corrosion, or other damage.

If the seal is damaged, it **must** be replaced. Refer to Procedure 008-016 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-016-tr.html)



Inspect the thermostat for damage.



Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

Do **not** allow the thermostat or thermometer to touch the container.

Heat the water and check the thermostat as follows:

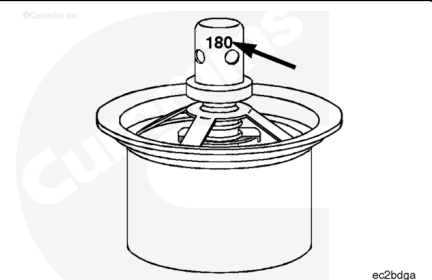


The nominal operating temperature is stamped on the thermostat.

- Thermostat **must** begin to open within 1°C [2°F] of the nominal temperature.
- Thermostat **must** be fully open within 12°C [22°F] of the nominal temperature.

The fully open distance between the thermostat flange and housing is 11 mm [0.435 in].

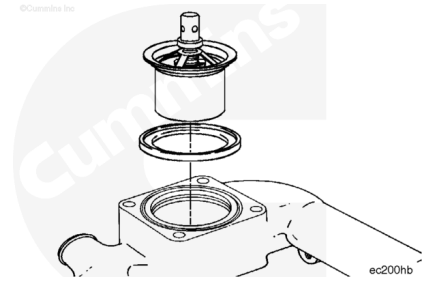
If the thermostat operates properly and more than 100 cc [3.3 fl oz] leakage is detected during in-chassis test, replace the thermostat. Refer to Procedure 008-016 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-016-tr.html)



Install

Install the thermostat in the housing.

Install a new o-ring seal in the groove on the thermostat housing mounting surface.

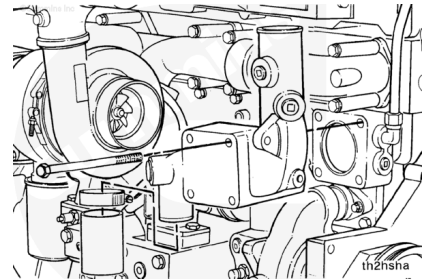


Install the hose on the thermostat housing bypass outlet.

Install the thermostat housing and four mounting capscrews.

Tighten the mounting capscrews.

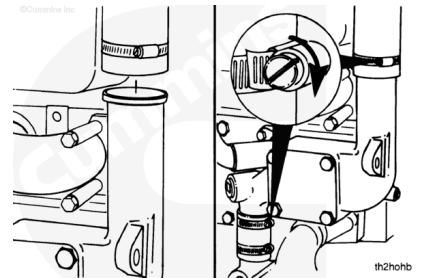
Torque Value: 54 n•m [40 ft-lb]



Install the upper coolant hose to the thermostat housing outlet.

Tighten the coolant bypass hose clamps.

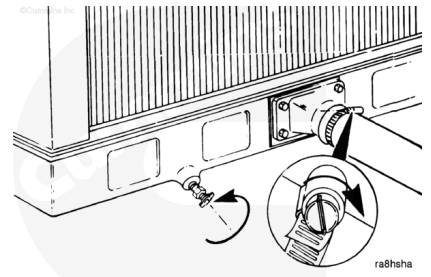
Torque Value: 5 n•m [44 in-lb]



Close the cooling system draincock, and install the lower coolant hose.

Tighten the hose clamp.

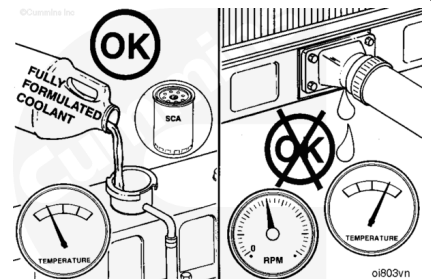
Refer to the OEM's specifications for the correct torque value.



The correct concentration of coolant additives **must** be used in the cooling system. Refer to Procedure 018-004 in Section V (</qs3/pubsys2/xml/en/procedures/101/101-018-004-om-auto.html>) of the Operation and Maintenance Manual, ISM and ISMe, Bulletin 3666319.

Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)

Fill the cooling system. Refer to Procedure 008-018 in Section 6. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-om-ind.html>)



Operate the engine until it reaches 80°C [180°F], and check for coolant leaks.

Last Modified: 19-Dec-2011
